



EC-220 Computer System Architecture

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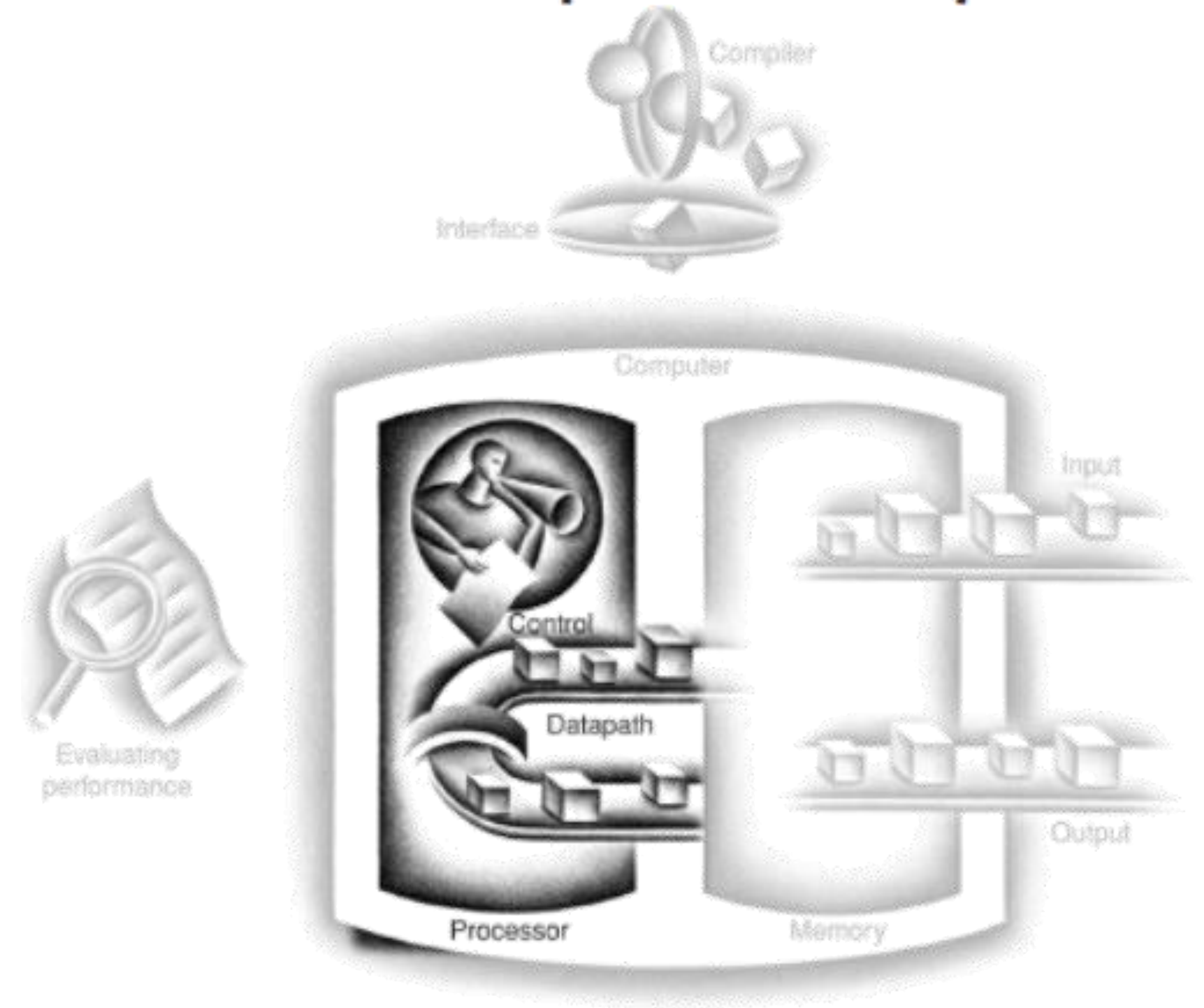
National University of Science and Technology (NUST)

Week # 11_OLE

The Processor

- **Performance of Computer**
 - Instruction Count-IC
 - CPI
 - Clock Cycle Time
 - **Compiler and ISA determine IC**
 - **Implementation of Processor determines both the CPI and Clock Cycle Time**
- Single Cycle and Multi-Cycle / Pipelined Processor

Five Classic Components of a Computer



MIPS Implementation Coverage

- **Implementation of a subset of Core MIPS Instr. set**
 - Memory-reference (lw, sw)
 - Arithmetic-logical (add, sub, AND, OR, Slt)
 - Branch and jump (beq, j)
- **Excluding Floating point and some integer instrs.**
 - Multiply, divide, shift etc.
- **Principles**
 - To Create a datapath
 - Designing of control

Overview of Implementation

- **For every instr. execution, the first two steps are same**
 - Fetch the instr. - Using PC value to read instr. memory
 - Decode the instr. – Reading one or Two Registers
- **Execution Step**
 - Among three instr. Classes, the majority of actions are similar
 - E.g. all instr. Classes except jump, use the ALU after reading the registers.
 - Memory-reference instrs. For address calculation
 - Arithmetic-logical instrs. For operation execution
 - branches for comparison
- **After using the ALU, the actions required to complete various instrs. Differ**
 - i.e to access memory R/W, Data writing in Reg., updating PC and so on.

Basic MIPS Implementation

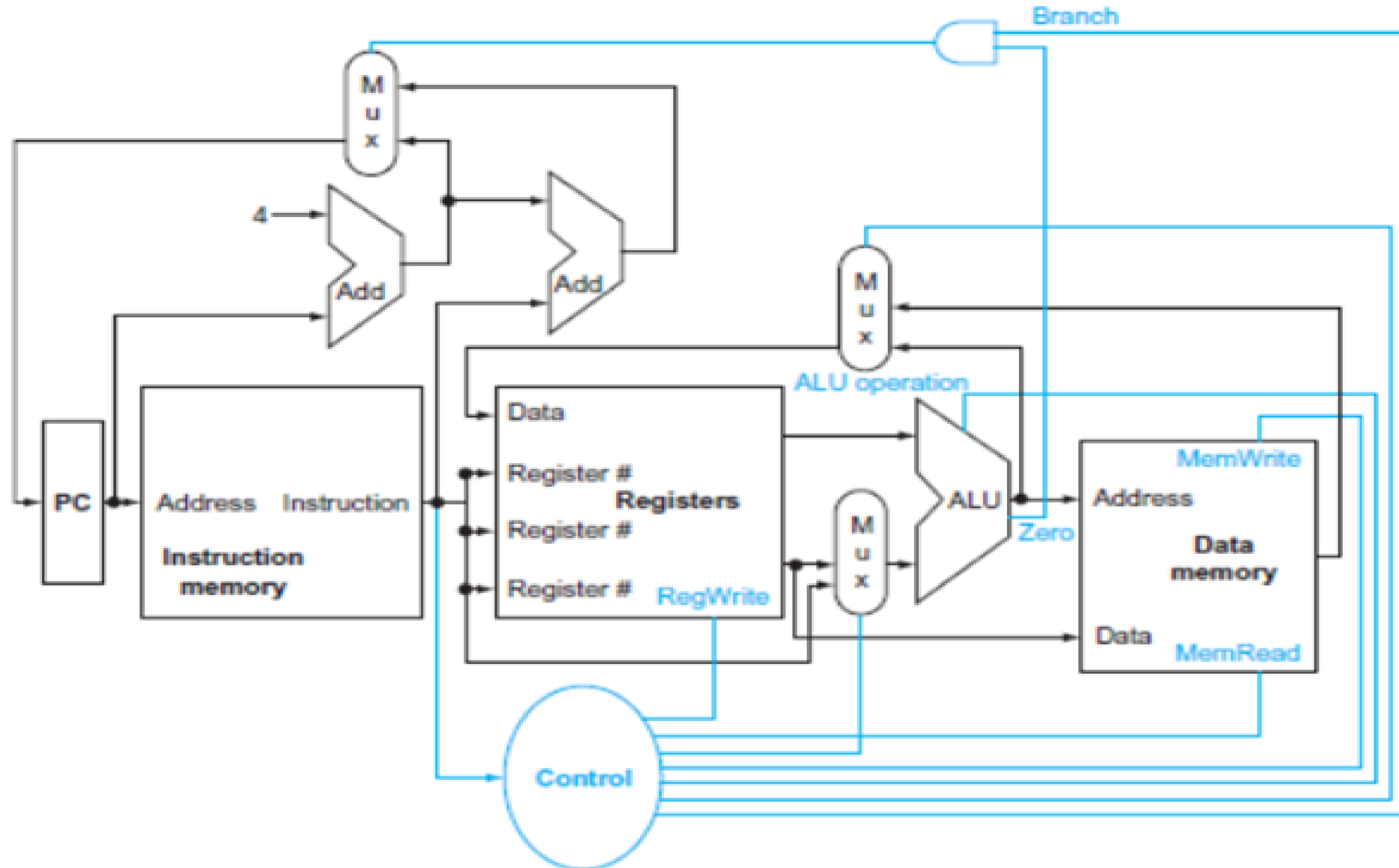
- **Data Path**

- Represents the flow of data through interconnection of functional units

- **Control unit**

- Takes instruction as an input and generates signals to control the functional units and datapath for proper execution of that instr.

Basic MIPS Implementation



Review: Logic Design

■ Combinational Element

- An operational element,
- such as AND gate or an ALU

■ State Element

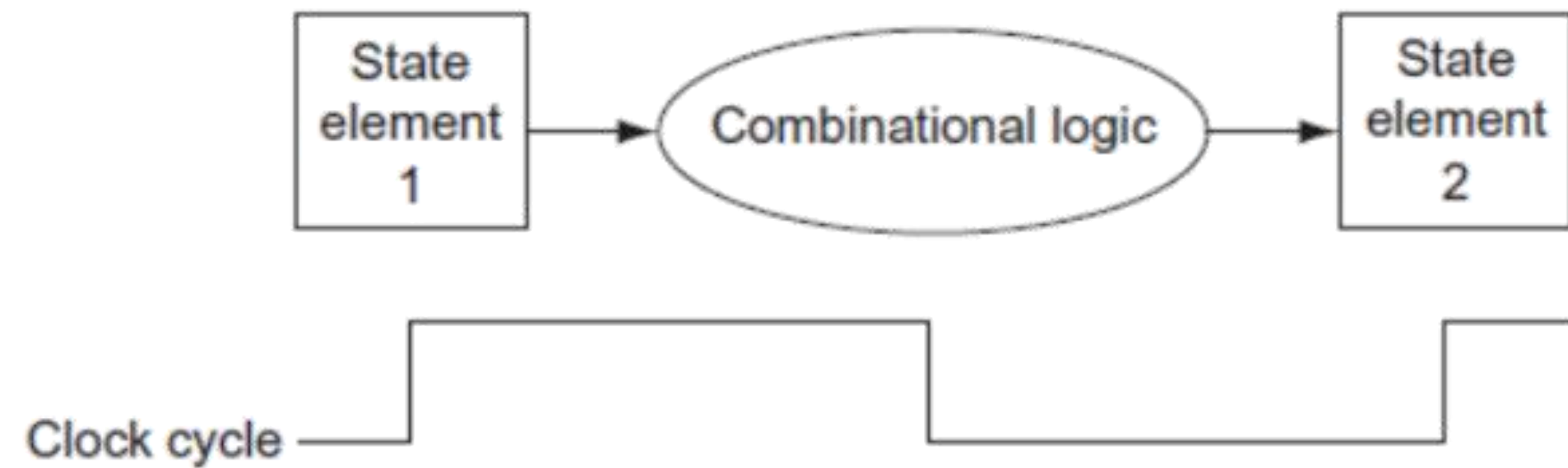
- A memory element, such as a register or a memory
- At least two inputs (data value, clock) e.g. Sequential circuits

■ Clocking Methodology

- The approach used to determine when data is valid and stable relative to the clock.

■ Edge-Triggered Clocking

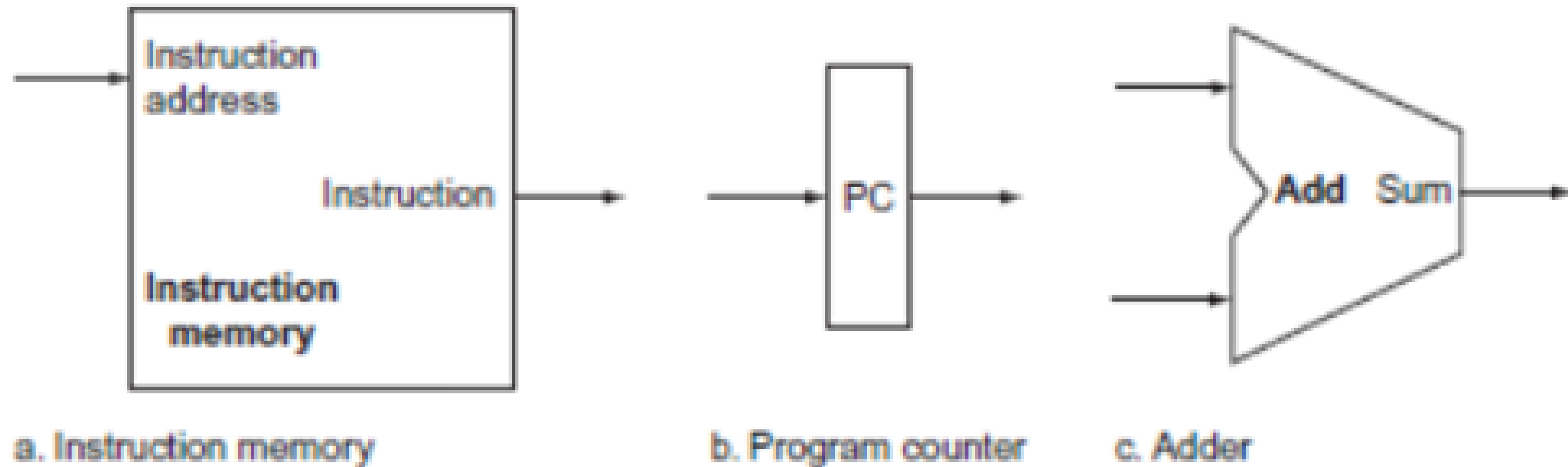
- A clocking scheme in which all state changes occur on a clock edge e.g. low to high or vice versa.



Building a Datapath – Datapath Elements

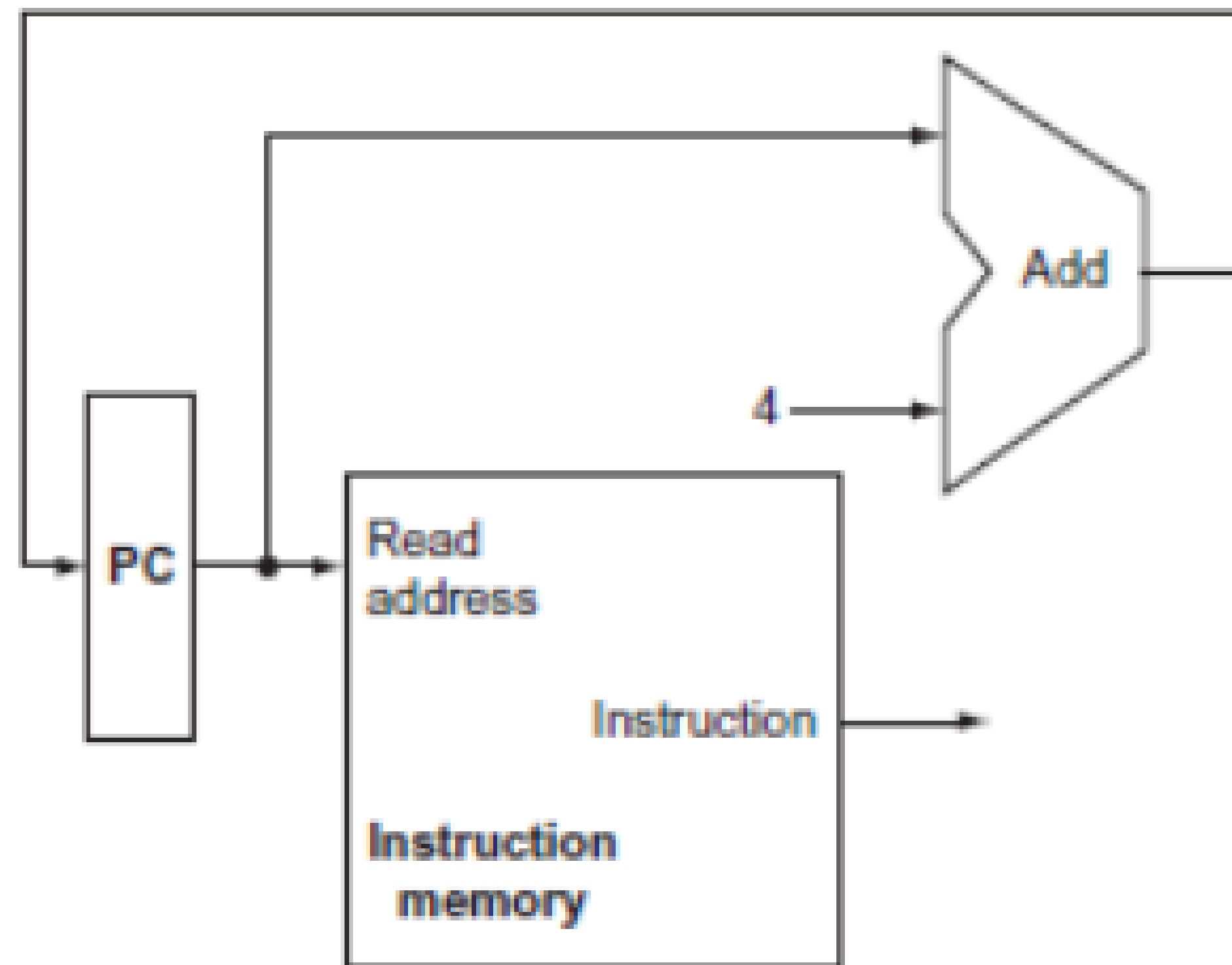
■ Datapath Element

- A unit used to operate on or hold data within a processor. In the MIPS implementation, the datapath elements include the instruction and data memories, the register file, the ALU, and adders.



Datapath Elements

- Datapath that fetches instructions and increments the PC to obtain the address of the next sequential instruction.



IE 4.6 A portion of the datapath used for fetching instructions and incrementing

Datapath Elements

■ Register File

- A state element that consists of a set of registers that can be read and written by supplying a register number to be accessed.

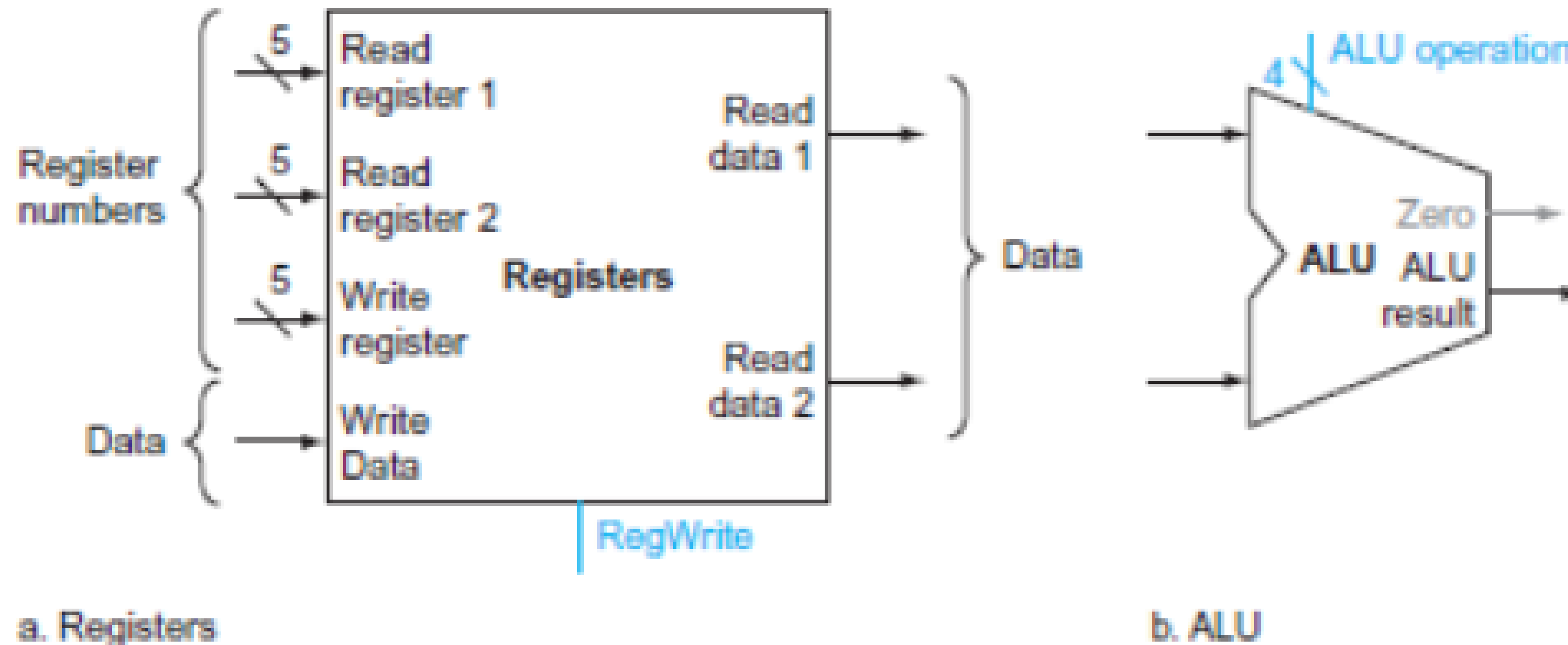


FIGURE 4.7 The two elements needed to implement R-format ALU operations are the

Datapath Elements

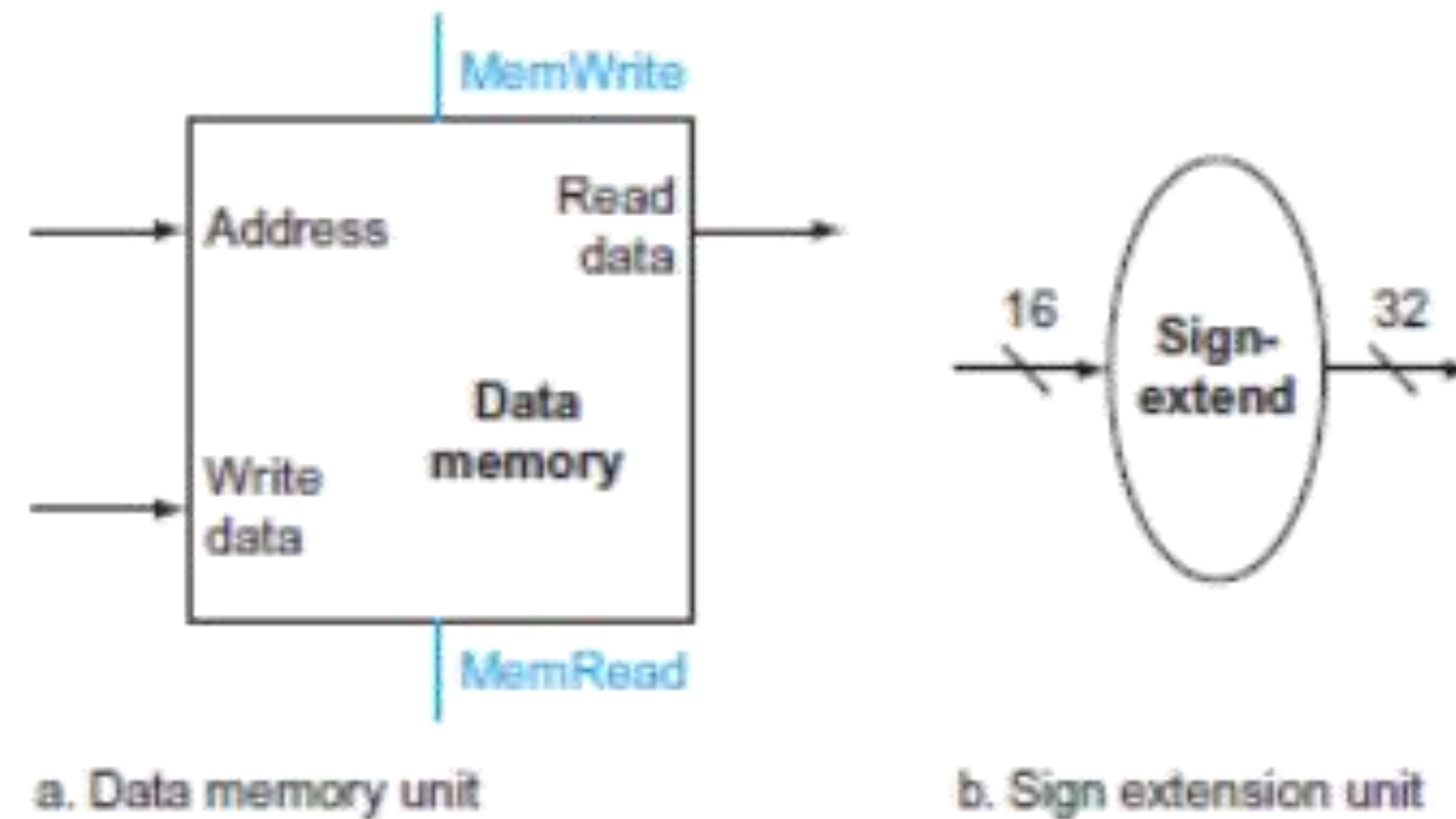


FIGURE 4.8 The two units needed to implement loads and stores, in addition to the

Branch Instrs. Implementation

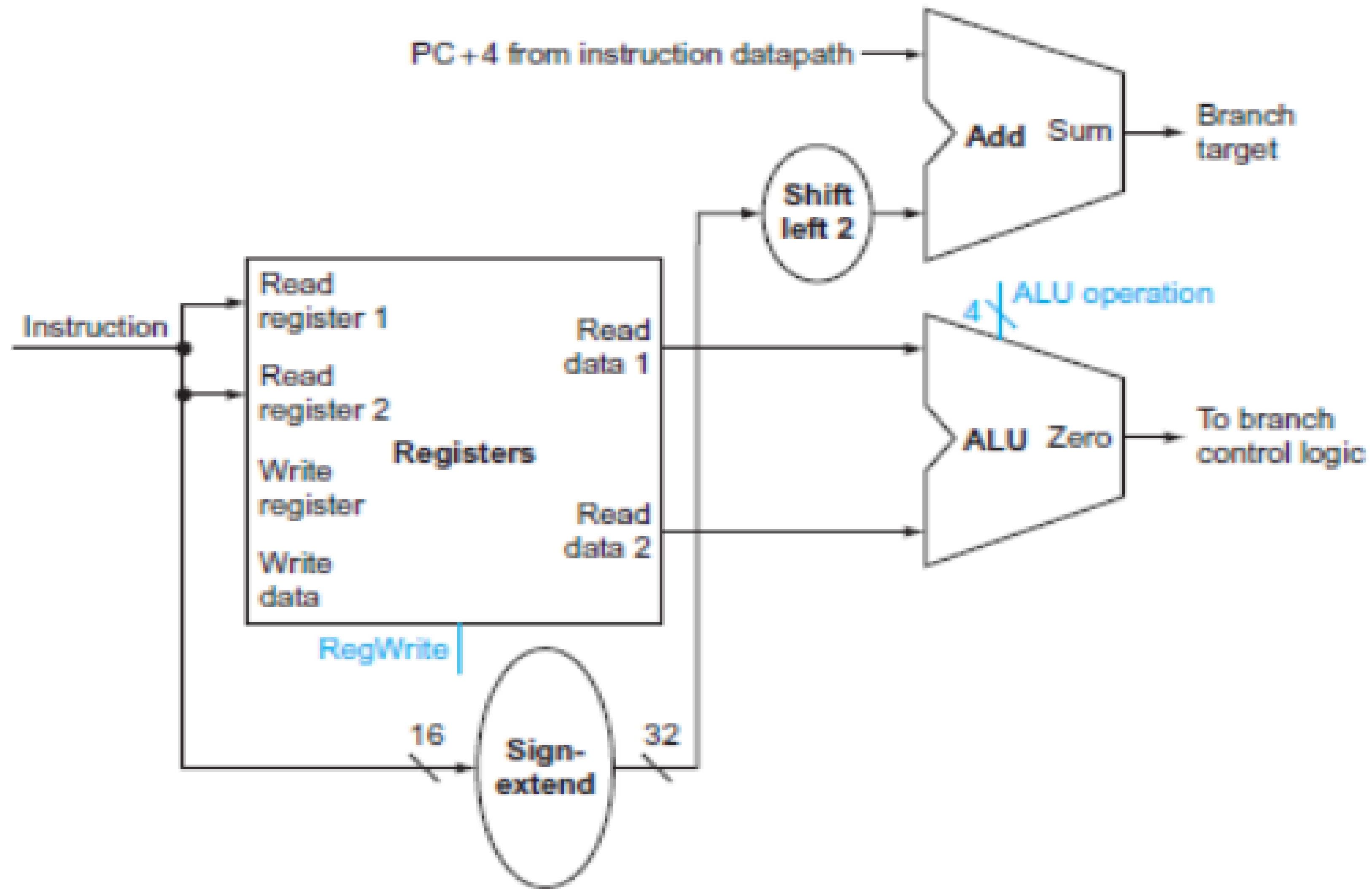
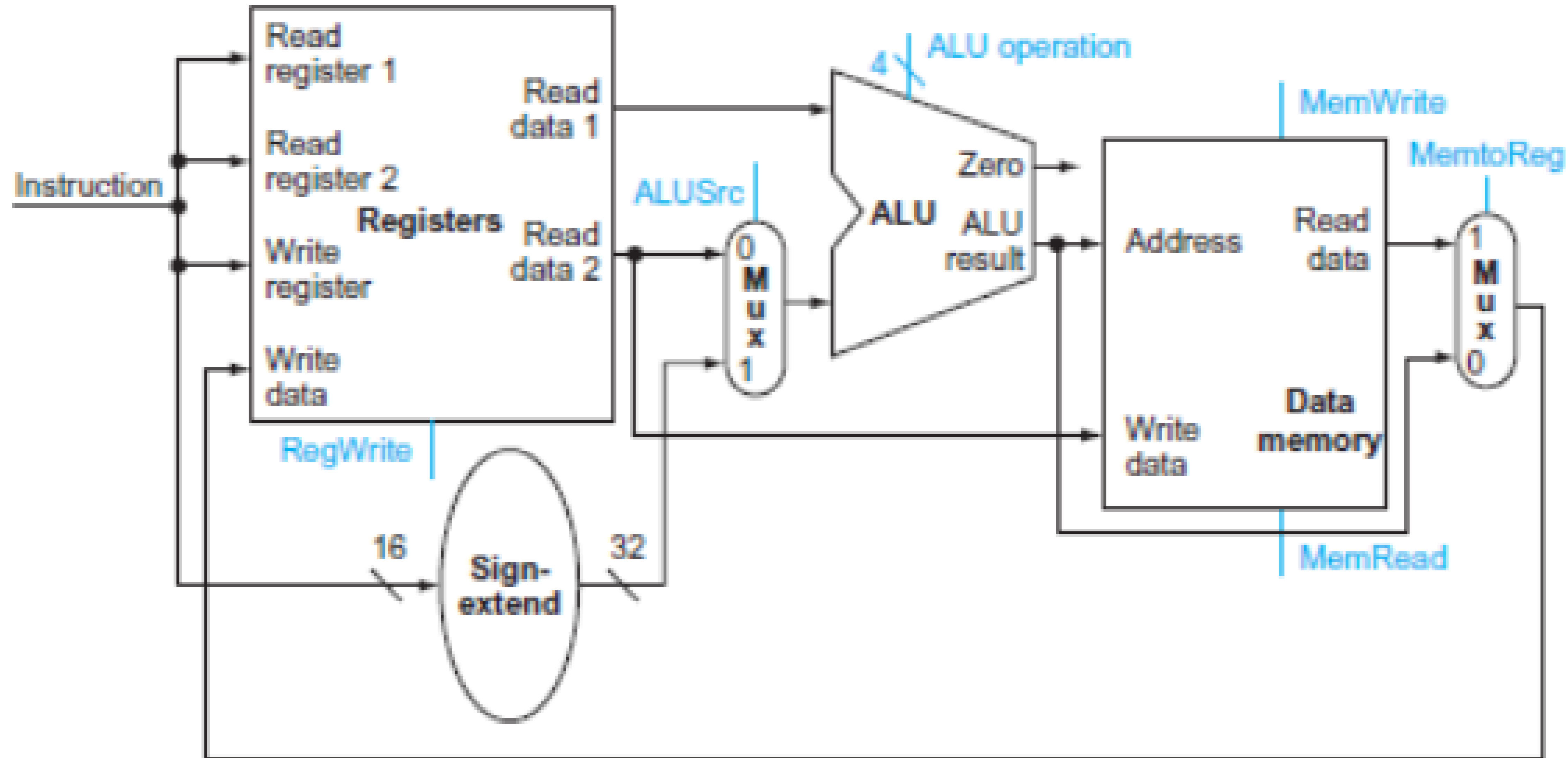


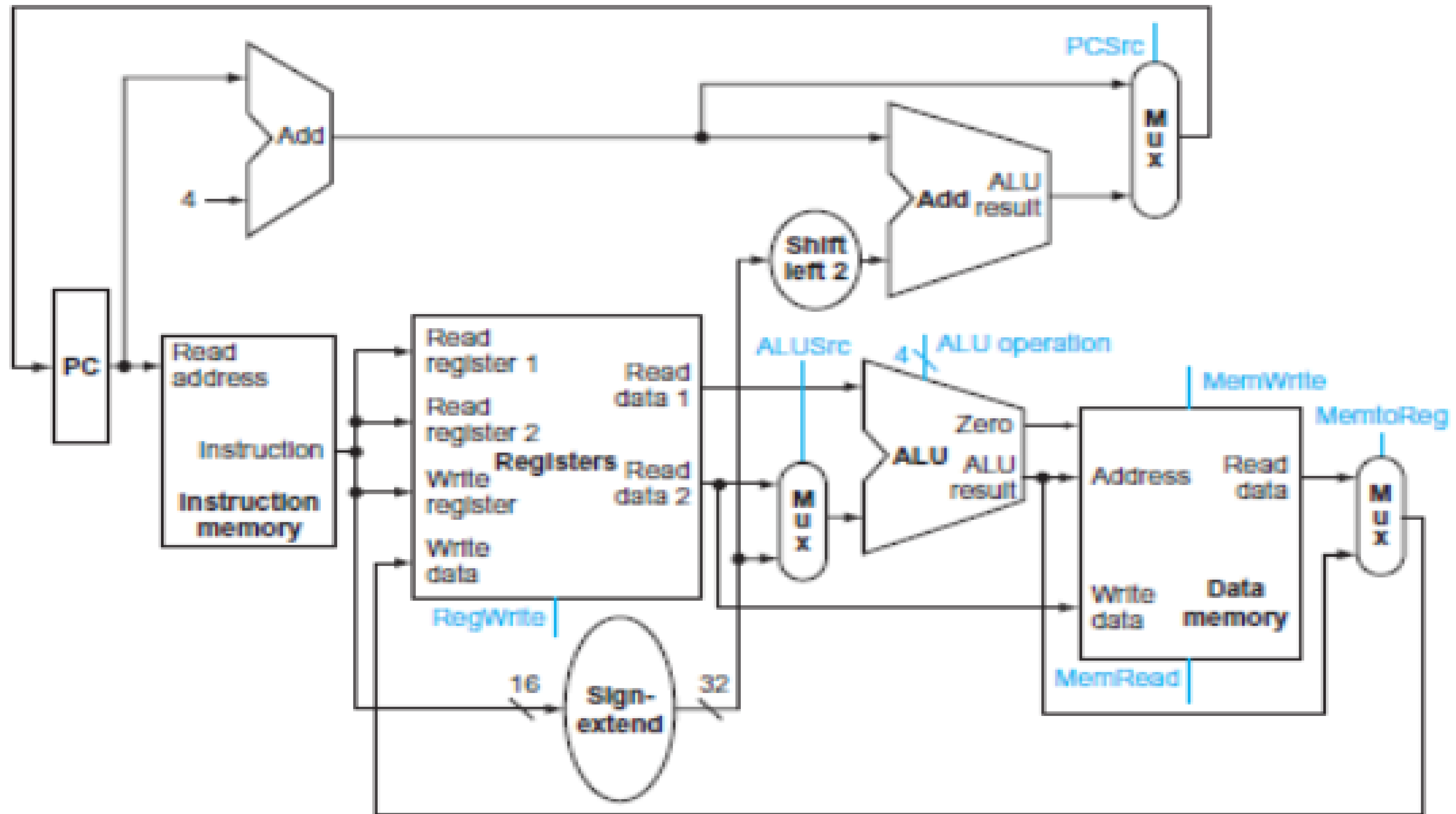
FIGURE 4.9 The datapath for a branch uses the ALU to evaluate the branch condition and

R- type and Mem-ref. instrs. **DataPath**



E 4.10 The datapath for the memory instructions and the R-type instructions. This example shows how a single

Complete DataPath of MIPS Integer Core Subset



4.11 The simple datapath for the core MIPS architecture combines the elements required by d